

Jihwan Choi

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Profile

Software engineer building and operating **production Generative AI** and the MLOps inference/evaluation platform behind autonomous-driving AI. I ship multimodal **Gemini on Vertex AI** (live in customer releases), LLM cost/usage observability, RAG, and agentic tooling — on top of a shared Kubernetes GPU cluster I co-designed (scheduler & queue policy) and the **SVEvalFlow** workflow I own end to end (Inferit, MTBF Validation, PRISM). I turn engineering into measurable business value: a 20,000-hour validation cut from 6+ months to 1.5 months (430,851 clips at 99.87%), a cost-driven on-premise move saving ₩448M (55%) vs AWS, and a log-analysis platform over 80M daily events preventing ~₩100M/month in overbilling.

Experience

Stradvision, Software Engineer

Seoul, South Korea
July 2022 – Present

MLOps Platform [\[Link\]](#)

Shared GPU Cluster Co-Design & SVEvalFlow (Inference/Eval) Owner (Oct 2025 – Present)

Tech: RKE2, Ray, KubeRay, KServe, Knative, MLflow, Airflow, ArgoCD, Harbor, lakeFS, GitHub Actions, Helm

- Co-designed and built a shared GPU cluster on RKE2 Kubernetes (60+ heterogeneous GPUs: RTX 3090/4080/4090/5090, RTX 6000), **leading scheduler & queue policy design** and integrating 15+ MLOps components (Ray/KubeRay, MLflow, Airflow, ArgoCD, KServe, Harbor)
- Collaborate on the training workflow (SVDeepFlow) and **solely own the inference/evaluation workflow (SVEvalFlow)** — Inferit, MTBF, and PRISM below
- Built 3 CI/CD pipelines (GitHub Actions + ArgoCD) and Scale-to-Zero serverless inference (KServe/Knative)
- Drove internal adoption of AI-native development tooling (Claude Code, Gemini CLI, Codex; MCP/Plugin/Skill) through hands-on engineering sessions and n8n automation prototypes

Inferit [\[Link\]](#)

svnet3 Build/Inference Acceleration & Artifact Management

Tech: FastAPI, Airflow, Kubernetes, lakeFS, Docker, PostgreSQL, HCP Object Storage

- Built version/config/artifact management for building svnet3 (in-house autonomous-driving vision model) across many configs
- Accelerated single-machine inference into distributed/parallel execution across cluster GPUs — cut a 20,000-hour validation from an estimated 6+ months to 1.5 months
- Versioned customer option files (e.g., camera parameters), removing the 1–2 days of engineer-PM search and hand-off each ticket previously required
- Supports 5+ major release versions, with per-ticket automated inference across OD/LD/TSR/TLR features (run multiple times a week) and integrated Evaluation-Driven Development (eval automation)

MTBF Validation [\[Link\]](#)

Large-Scale svnet3 Reliability Validation

Tech: FastAPI, Airflow, Kubernetes, Docker, PostgreSQL, HCP Object Storage

- Automated svnet3 inference on customer-ODD data before/after each release to flag false positives and AEB events. Processed 430,851 clips at 99.87% success (15,233h), with 205% higher throughput and 67.3% lower processing time vs a single machine — ~10 real false positives across 15,233h (~1 per 5,000h)
- Proposed and executed a cost-driven Cloud to On-Premise transition: AWS ₩820M to On-Premise ₩370M (55% saved), with GPU hardware benchmarks & spec proposals and HCP Object Storage
- Automation let one person operate validation previously feasible only at small scale. Visual Inspection UI plus **Gemini 2.5 Flash on Vertex AI** auto-FP assist (multimodal, structured output, human-in-the-loop)

PRISM [\[Link\]](#)

Customer Jira-Integrated ISP Conversion & Re-simulation Pipeline (Nov 2025 –)

Tech: FastAPI, Airflow, PostgreSQL, AWS (SQS, Lambda), Terraform

- **Phase 1 (pre-deployment):** Event-driven ISP conversion via Jira Webhook to SQS to Lambda, integrated with the in-house ISP tool. Connectivity test complete, with full deployment to follow Inferit. Designed for ~150 requests/month, ~80% faster than manual
- **Phase 2 (Resim, in dev):** Built the initial version and now act as technical lead, guiding another org's developer on the implementation. Dual-version inference (issue + latest) via SWIT Docker Images, Airflow DAG trigger

Data & Labeling Automation (2022 – 2025)

- **WLS:** Analyzed 80M daily labeling-log events to prevent ~₩100M/month overbilling (settlement-audit confirmed). Pinpointed bottlenecks feeding UI/UX and the ALAS model (+20% labeling efficiency) (Django, MongoDB, AWS SQS)
- **ALAS:** Auto-labeling microservice, reduced manual annotation workload by ~35% (FastAPI, AWS SQS, AWS SNS, AWS ECS/EKS, CDK)
- **IGS:** External-customer data ingestion service planning & architecture design (design-only) (Django, Airflow, Kubernetes)
- **SURF Compass:** Device monitoring service planning & design (design-only) (FastAPI, PostgreSQL)
- **RNS:** Automated release-note generation, 50% faster authoring (4h to 2h) (NestJS, Atlassian API)
- **SLT:** 2D LD labeling tool, improved productivity by 30% (C++/MFC)

Korea Institute of Robotics & Technology Convergence, Researcher

Pohang, South Korea

- Developed real-time gesture-based Human-Robot Interaction system
- Supported autism diagnosis assistance robot project

Jan 2022 – July 2022

Skills

Languages: Python, C++, TypeScript

Backend: Django, FastAPI, NestJS

Database: PostgreSQL, MongoDB, Redis

Cloud: GCP (Vertex AI), AWS (EC2, EKS, ECS, SQS, SNS, Lambda, S3, Bedrock)

Infrastructure: Kubernetes (RKE2), Docker, Terraform, CDK, Rancher

ML/MLOps: Ray, KubeRay, MLflow, KServe, Knative, lakeFS

Workflow: Airflow, Celery

CI/CD: ArgoCD, GitHub Actions, Harbor, Helm

Monitoring: Grafana, Prometheus, Loki, DataDog

Generative AI / LLM: Vertex AI (Gemini), AWS Bedrock, multimodal, structured output, RAG / embeddings, prompt engineering, LLM cost & observability, agentic / tool-use, multi-model orchestration, Claude Code / MCP / Skills

Tools: Git, Jira, Confluence, Atlassian API

Generative AI Engineering

- **Gemini on Vertex AI (production, multimodal):** Gemini 2.5 Flash for auto-FP detection of AD vision results (image + structured metadata). JSON-schema structured output, per-request token tracking, batched async + dedup (~40% fewer calls), human-in-the-loop. Live in real customer-release evaluation (MTBF)
- **LLM cost & usage observability (AWS Bedrock):** Token-level, per-model/per-team cost-per-request attribution via CloudWatch + Slack, Terraform IaC. 90%+ accuracy vs billing, adopted as an org cost-optimization tool
- **RAG knowledge system (LLM Wiki):** Hybrid keyword + embedding retrieval (text-embedding-3-small, cosine top-k) → grounded answers with citations, freshness, and uncertainty. CI-rebuilt index (Git + Markdown)
- **Agentic LLM tooling:** NL intent → cluster diagnostics/storage tools (Claude Code Skills), multi-step orchestration + guardrails (dry-run, rollback, gates). Multi-model self-critique across Claude/GPT/Gemini (jhwanchoi/dotfiles)
- **Tech:** Vertex AI (Gemini), AWS Bedrock, embeddings/RAG, Claude Code, Gemini CLI, Codex, MCP, CloudWatch, Terraform, n8n

Education

BS **Handong Global University**, Electronic Engineering & Computer Science —
GPA 3.3/4.5

Feb 2018 – Feb 2022

Coursework: Data Structure, Algorithm, Computer Architecture, Deep Learning, Machine Learning